Report for Computer Simulation of Physical Systems I

Lennard-Jones Liquid: Structural and Dynamical Properties

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1 Introduction

This report explores the physical properties of a Lennard-Jones liquid, modeling liquid argon as in Rahman [1]. The main goal is to calculate structural and dynamical observables using different simulation techniques presented in the course: NVE molecular dynamics, Nosé–Hoover molecular dynamics, and Monte Carlo simulations.

We focus on the following physical quantities: the pair correlation function $g(r)$, the structure factor $S(k)$, the diffusion coefficient D, and the average potential energy U. For each quantity we collect data using the specified simulation methods, present the results, discuss the physical interpretation, and compare with the results of Ref. [1].

2 Simulation Setup

All simulations concern liquid argon with Lennard-Jones parameters $\varepsilon/k_{\mathrm{B}} = 120\,\mathrm{K}$ and $\sigma = 3.4\,$Å, as in Rahman [1]. Simulations are performed at temperature $T = 94.4\,\mathrm{K}$ and density $\rho = 1.374\,\mathrm{g\,cm^{-3}}$ using $N = 864$ particles and a time step $\Delta t = 0.01\,\mathrm{ps}$.

2.1 Lennard–Jones Model

We simulate a single–component fluid interacting via the Lennard–Jones (LJ) pair potential

$$V(r) = 4\varepsilon\left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^{6}\right],$$

with cubic periodic boundary conditions. Distances are computed with the minimum–image convention and interactions are truncated at a cutoff $r_{\mathrm{cut}} \leq L/2$; long–range contributions beyond r_{cut} are accounted for by tail corrections as implemented in each code.

2.2 Molecular Dynamics (NVE and Nosé–Hoover)

Equations of motion are integrated with velocity–Verlet using the timestep given before. After a short NVT equilibration, production is run either: (i) in the microcanonical ensemble (NVE), conserving total energy and sampling equilibrium configurations at fixed (N, V, E); or (ii) in the canonical ensemble (Nosé–Hoover thermostat with weak coupling), sampling at fixed (N, V, T) while preserving the correct static structure. We remove the initial center–of–mass drift and use periodic boundaries throughout.

2.3 Monte Carlo Simulations

Independent equilibrium configurations at fixed (N, V, T) are generated via single–particle Metropolis moves: a particle is displaced by a random vector of maximal amplitude $d_{\max}$ and the trial configuration is accepted with probability $\min\{1, \exp[-\beta\Delta E]\}$. During equilibration (set to 10^4

cycles) we adapt $d_{\max}$ to target an acceptance ratio $\simeq 0.5$; production samples are then taken at regular intervals. Energies are evaluated with the same cutoff/tail scheme as in MD. Monte Carlo provides equilibrium static properties, while dynamical quantities are assessed from MD only.

3 Results and Discussion

3.1 Pair Correlation Function $g(r)$

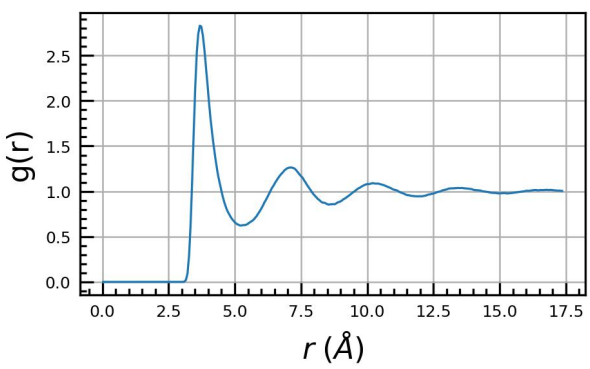
3.1.1 NVE Molecular Dynamics

In this subsection we compute the pair correlation function $g(r)$ from the NVE molecular dynamics trajectories. The radial distribution is estimated via histogramming interparticle distances, using $N_{bins} = 300$ and $\Delta r = (L/2)/N_{bins}$.

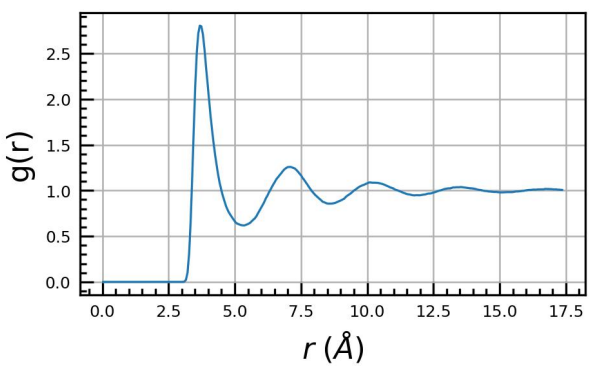
After NVT equilibration, we ran a microcanonical (NVE) production trajectory with periodic boundary conditions. At regular intervals we accumulated all pair distances using the minimum-image convention, binned them in r, and normalized the histogram by $4\pi r^2 \rho \, \Delta r$ (and the number of sampled configurations), yielding the radial distribution function $g(r)$ shown in figure 1a.

3.1.2 Nosé–Hoover Molecular Dynamics

We propagated the dynamics at fixed temperature using a Nosé–Hoover thermostat with periodic boundaries and a mass parameter $Q = 10$ (LJ units). This value sets a weak-to-moderate coupling, ensuring stable temperature control without overdamping the short-time dynamics. Pair distances were collected along the production segment exactly as in NVE (minimum-image, binning in r), and the same normalization $4\pi r^2 \rho \, \Delta r$ was applied (Plot in figure 1b). Since $g(r)$ is a static equilibrium observable, the NVT result should match NVE at the same (ρ, T).



(a) Pair correlation function $g(r)$ from NVE MD vs distance r



(b) Pair correlation function $g(r)$ from Nosé–Hoover MD vs distance r

Figure 1: Pair correlation function $g(r)$ obtained from NVE (a) and Nosé–Hoover (b) molecular dynamics at the same state point.

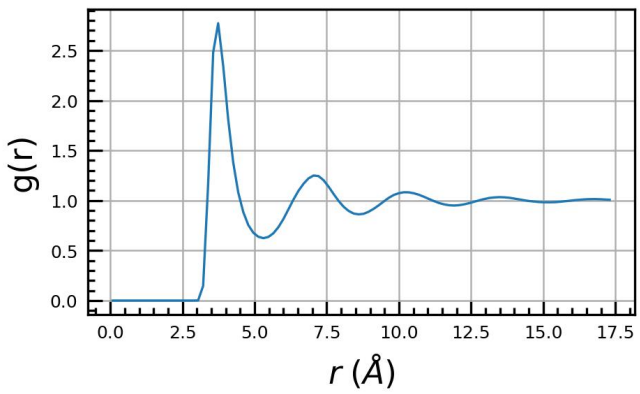


Figure 2: Pair correlation function $g(r)$ obtained from Monte Carlo simulations vs distance r.

3.1.3 Monte Carlo Simulation

Finally we estimate $g(r)$ from Monte Carlo simulations.
Configurations were sampled with Metropolis moves in the canonical ensemble at the target (ρ, T). For each accepted configuration we computed all pair distances with the minimum-image convention, accumulated the histogram in r, and applied the standard normalization $4\pi r^2 \rho\, \Delta r$ to obtain $g(r)$.

3.1.4 Discussion

The three $g(r)$ estimates overlap within plotting resolution. All display the canonical Lennard–Jones liquid structure also reported in Rahman: a sharp first-neighbour peak, a subsequent minimum, damped oscillations, and convergence to $g(r) \to 1$ at large r. The peak position and the overall damping pattern are consistent with the reference trend once distances are expressed in Angstrom units. No systematic shift of peak positions or baseline is observed between ensembles. The Monte Carlo estimate, which samples the equilibrium distribution directly, also coincides with MD. Any extremely tiny differences between the curves are compatible with finite statistics and finite-size effects, rather than with algorithmic bias.

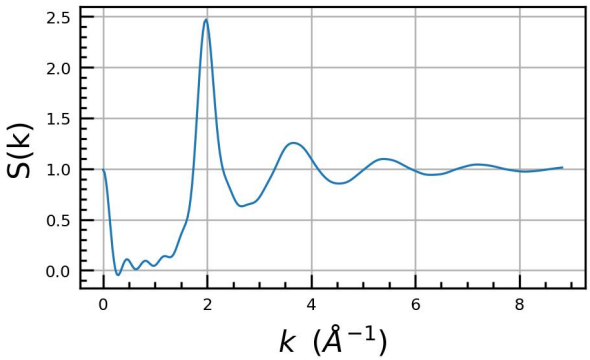
3.2 Structure Factor $S(k)$

3.2.1 Fourier Transform of $g(r)$ (NVE MD)

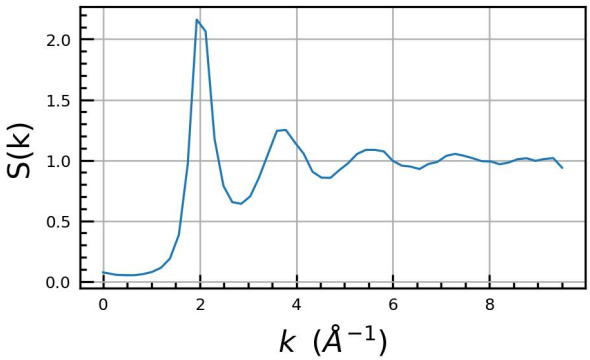
The structure factor $S(k)$ is first obtained by numerically Fourier-transforming $g(r)$ computed from the NVE simulations, using the relation $S(k) = 1 + 4\pi\rho\int_0^\infty [g(r) - 1]\sin(kr)/(kr) r^2\, dr$ and evaluating the integral on a uniform k grid of $n_k = 300$ points up to $k_{max} = 32$ (LJ). For very small k we replace $\sin(kr)/(kr)$ by its Taylor series to prevent numerical instabilities.

3.2.2 Direct Sampling Method (NVE MD)

We also compute $S(k)$ directly from the particle positions using its general definition $S(\mathbf{k}) = \frac{1}{N}\langle \sum_{i,j} e^{-i\mathbf{k}\cdot(\mathbf{r}_j - \mathbf{r}_i)} \rangle$ for $\mathbf{k} \neq 0$. Practically, we accumulate $\sum_{i,j} e^{-i\mathbf{k}\cdot(\mathbf{r}_j - \mathbf{r}_i)}$ for each configuration during the time evolution, then average over these values and divide by the number of particles. Because the liquid is isotropic, $S(\mathbf{k})$ depends only on the magnitude $k = |\mathbf{k}|$. Therefore, we group wavevectors into k-shells (by modulus) and average over all directions within each shell to obtain the radial function $S(k)$.



(a) $S(k)$ from Fourier transform of $g(r)$ vs wavenumber k



(b) $S(k)$ from direct sampling vs wavenumber k

Figure 3: Static structure factor $S(k)$ obtained from (a) the Fourier transform of $g(r)$ and (b) the direct sampling method.

3.2.3 Discussion

In Rahman's paper the peaks of $S(k)$, obtained through Fourier transform, agree with our results for both methods. The structure factor obtained from the Fourier transform of $g(r)$ appears smoother because the calculation of $g(r)$ involves an integration over r and a consequent evaluation on a dense k-grid. In contrast, the direct-sampling curve is noisier because it uses only the discrete wavevectors of the box and averages over a limited number of directions and configurations. On the other hand, at small wavevectors, the directly sampled $S(k)$ shows a smooth, almost parabolic behavior and approaches a finite small-k limit $S(k \to 0^+) < 1$, consistent with the Taylor expansion of $S(k)$ for $k \to 0$. In contrast, the $S(k)$ obtained from the Fourier transform of $g(r)$ exhibits a non-physical shape for $k \lesssim 5$, starting near 1, dropping rapidly and showing spurious oscillations before matching the direct result at larger k. These artifacts arise from the numerical evaluation of the Fourier integral, mainly due to the finite cutoff in r, which makes the FT-based method unreliable in the small-k regime.

3.3 Diffusion Coefficient D

3.3.1 Einstein Relation: Mean Square Displacement (NVE MD)

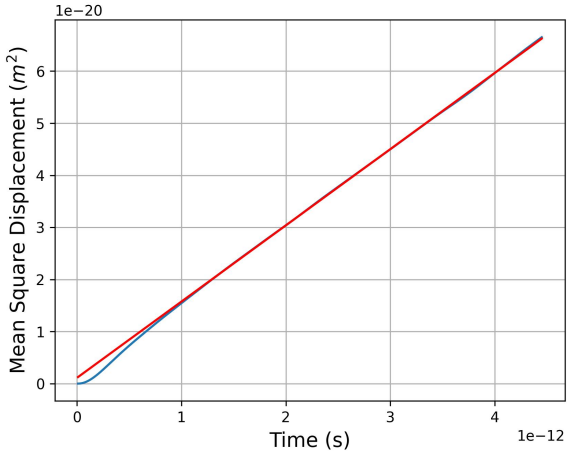
The diffusion coefficient D is first estimated from the long-time behavior of the mean square displacement (MSD), $\langle \Delta r^2(t) \rangle = \left\langle \left| \mathbf{r}(t) - \mathbf{r}(0) \right|^2 \right\rangle$, using Einstein's relation $D = \frac{\partial}{\partial t} \frac{\langle \Delta r^2(t) \rangle}{6}$.

We ran 20 simulations and for each we obtained the slope of the linear fit of the MSD vs time (Fig. 4a). After converting from LJ units to SI and averaging over all simulations, we found that $D = 2.427 \cdot 10^{-5} cm^2 s^{-1}$.

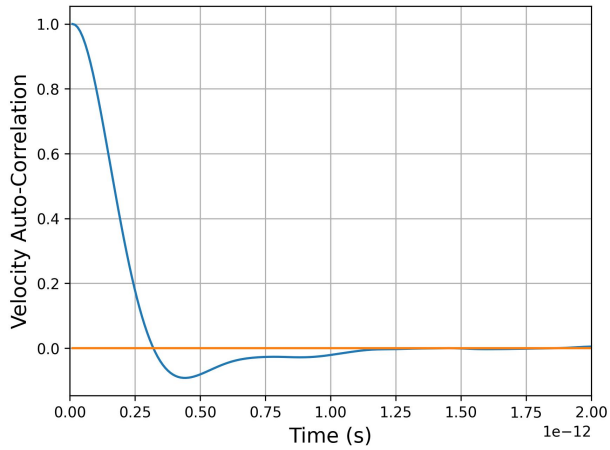
3.3.2 Green–Kubo Relation: Velocity Autocorrelation (NVE MD)

We also estimate D from the velocity–velocity autocorrelation function $C_{vv}(t)$ via the Green–Kubo relation $D = \int_0^\infty dt\, C_{vv}(t)$ with $C_{vv}(t) = \frac{1}{3} \langle \mathbf{v}(t) \cdot \mathbf{v}(0) \rangle$.

We took data from the same simulations used to estimate D with Einstein's relation. We first computed the velocity autocorrelation function $C_{vv}(t)$ (not normalized) and integrated it over the full simulation window to obtain D (In principle, one may truncate the integral near the first zero crossing to reduce noise). After unit conversion and averaging over runs, we obtained $D = 2.498 \cdot 10^{-5} cm^2 s^{-1}$. Moreover, we plot the normalized velocity autocorrelation function vs time to observe its behavior (Figure 4b), in particular to visualize its decay and the time it reaches the 0 plateau.



(a) Mean square displacement $\langle \Delta r^2(t) \rangle$ as a function of time t from NVE molecular dynamics, together with a linear fit in the diffusive regime used to extract D.



(b) Normalized velocity autocorrelation function from NVE molecular dynamics vs time t.

Figure 4: Plots produced with the data of *one* of the simulations used to compute D.

3.3.3 Discussion

From our estimations, we observe that Einstein's relation is more consistent with Rahman's one ($D = 2.43 \cdot 10^{-5} cm^2 s^{-1}$), but also the Green-Kubo relation gives a close result. In particular, their statistical errors, obtained through the variance of the uncorrelated average of D from different simulations, are $\sigma_D = \pm\ 0.05 \cdot 10^{-5} cm^2 s^{-1}$ for Einstein's relation method and $\sigma_D = \pm\ 0.09 \cdot 10^{-5} cm^2 s^{-1}$ for Green-Kubo one.

At very short times the motion is ballistic: velocities remain strongly correlated with their initial values, so the velocity autocorrelation function (VACF) is positive and the mean square displacement (MSD) is not linear. As collisions and intermolecular interactions accumulate, the system progressively loses memory of the initial velocities; the VACF decays toward zero (showing a small negative tail due to collision backscattering), and in the same time window the MSD crosses over to a linear trend. This marks the onset of diffusive dynamics, where the diffusion coefficient is well defined and consistent estimates are obtained both from the slope of the MSD and from velocity autocorrelation function.

Analyzing pros and cons of both approaches, we can say that the MSD slope is easy to compute and does not need any extra constraint to be set, but a linear behavior must appear in

the plot, which is not always guaranteed for more complex systems. On the other hand, Green-Kubo relation needs a time cut off to be set for the integral. This affects the result, adding up to the integration error, but it can always be computed.

3.4 Average Potential Energy U

3.4.1 Monte Carlo Estimate of Average Potential Energy U

We estimate the average potential energy *per particle* using Metropolis Monte Carlo in the canonical ensemble at the target (ρ, T) in Lennard–Jones (LJ) units. After equilibration (with the maximum displacement tuned to an acceptance ratio $\simeq 0.5$), production configurations are sampled at a fixed stride. For each sampled configuration we evaluate the Lennard–Jones energy with periodic boundary conditions, minimum-image convention and cutoff r_{cut}; the standard long-range tail correction (assuming $g(r) \approx 1$ for $r > r_{\text{cut}}$) is then added *per particle*. The final estimate is the average of these per-sample values and in figure 5 we can see how they distribute around the mean and how it becomes stable after the equilibration cycles.

Result.

$$\frac{U}{N}\,(\text{LJ}) = -5.78324 \qquad \Rightarrow \qquad \frac{U}{N}\,(\text{SI}) = -9.58155 \times 10^{-21}\ \text{J}.$$

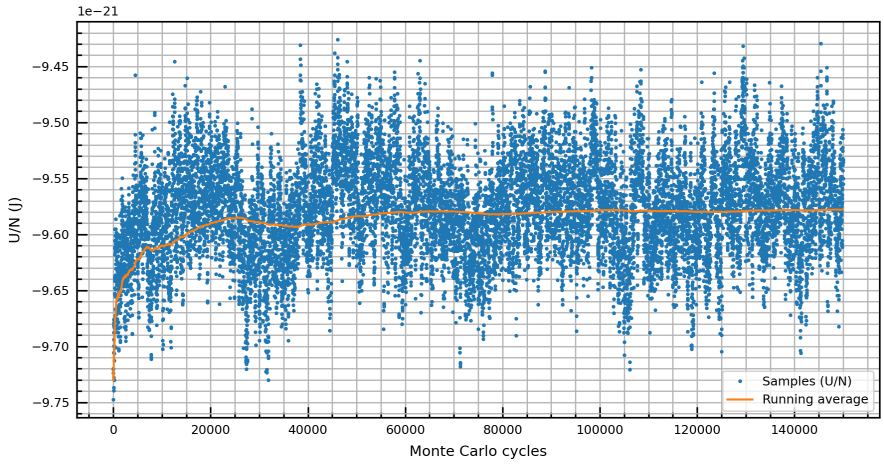
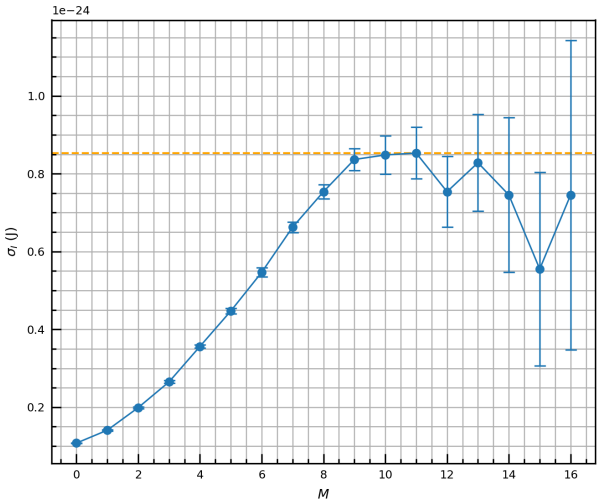


Figure 5: Series (Monte Carlo steps) of the potential energy U per particle from the Monte Carlo simulation, illustrating equilibration and fluctuations around the average value.

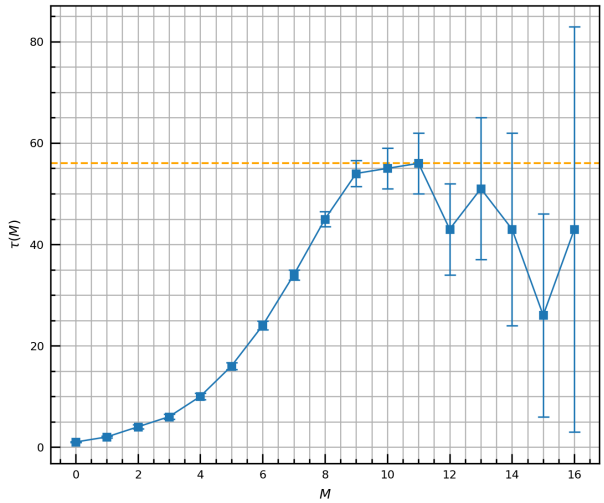
3.4.2 Blocking Analysis and Error ΔU

To quantify the statistical uncertainty due to time correlations, we apply a standard blocking analysis to the series of per-sample values of U/N. The left panel reports the standard error of the mean versus the blocking level M; the right panel shows the corresponding blocked correlation measure $\tau(M)$. The statistical error ΔU is read from the plateau of the left curve and corresponds to $\Delta U = \pm 8.53 \cdot 10^{-25}\, J$. For this analysis, we ran 1.5 million cycles of MC algorithm, sampling with a frequency of 10, meaning that the number of samples is $1.5 \cdot 10^5$. The plateau

is not very visible because as soon as it starts to appear, also the standard deviation increases due to the small number of elements inside the corresponding block. In order to achieve better results, we need to increase N and so the number of cycles, but we could not push further due to computational time constraints on the server.



(a) Standard error of U/N (J) vs. blocking level M.



(b) Blocked correlation measure $\tau(M)$ vs. blocking level M.

Figure 6: Blocking analysis for the MC estimate of U/N. The error ΔU is chosen from the starting of the plateau region of panel (a).

3.4.3 Discussion

We computed $\sigma_I^{plateau}/\sigma_I(0)$ to check if it is almost equal to $\sqrt{2\tau}$ as known from theory, but noticed a discrepancy in the two values: $\sigma_I^{plateau}/\sigma_I(0) = 7.38 \neq 10.44 = \sqrt{2\tau}$. This could be because the plateau is not perfectly reached, so we should increase N as already discussed. Therefore, we also compute τ using its definition and the result is $\tau = 40.11$, which means $\sqrt{2\tau} = 8.96$. This confirms the fact that our blocking analysis needs more samples to be accurate enough, since it underestimates the ratio between the errors and overestimates the actual correlation time.

References

[1] A. Rahman, “Correlations in the Motion of Atoms in Liquid Argon,” *Phys. Rev.* **136**, A405–A411 (1964).